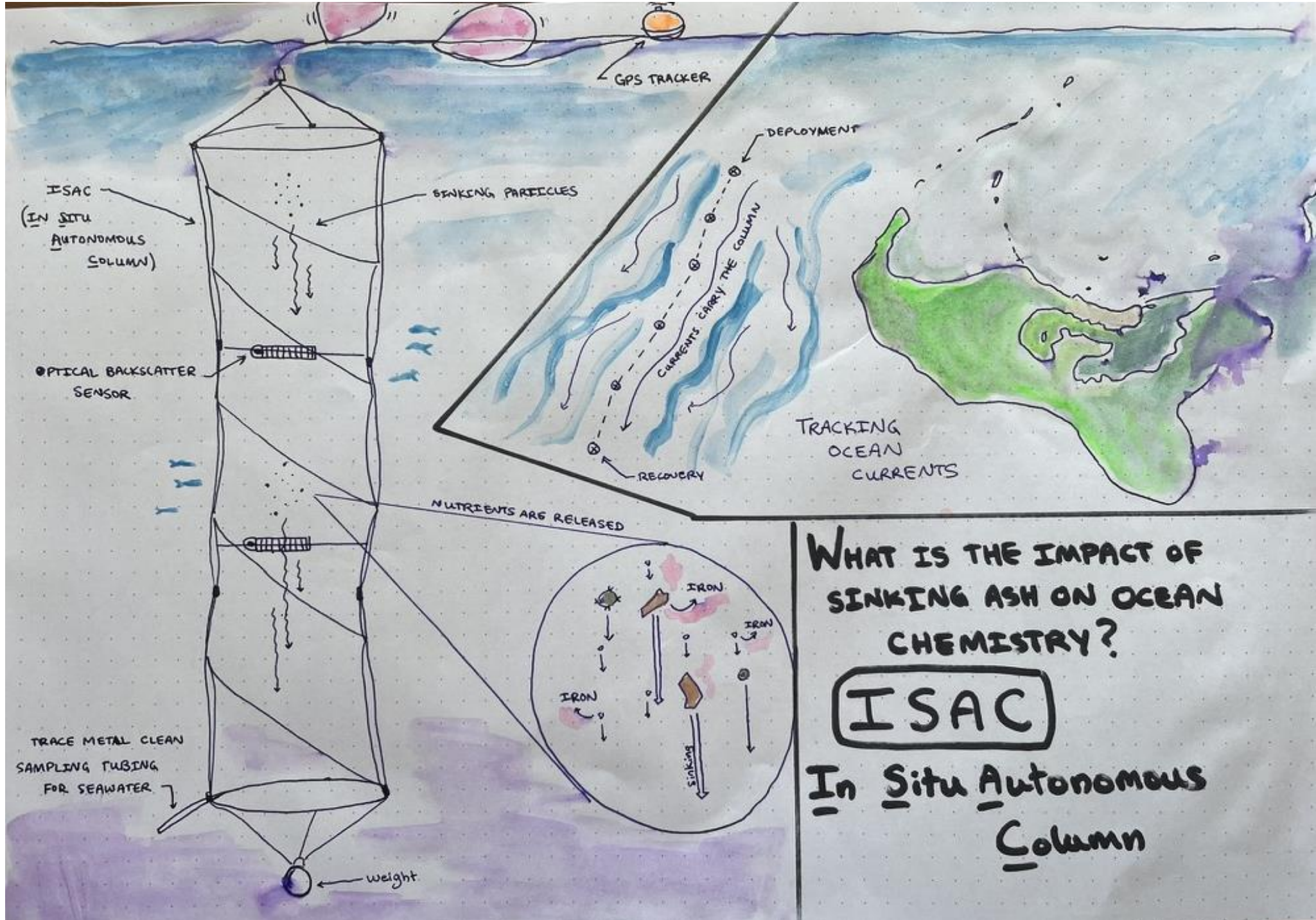


ISAC

Estimating sinking rates of particulate matter

Kiefer Forsch

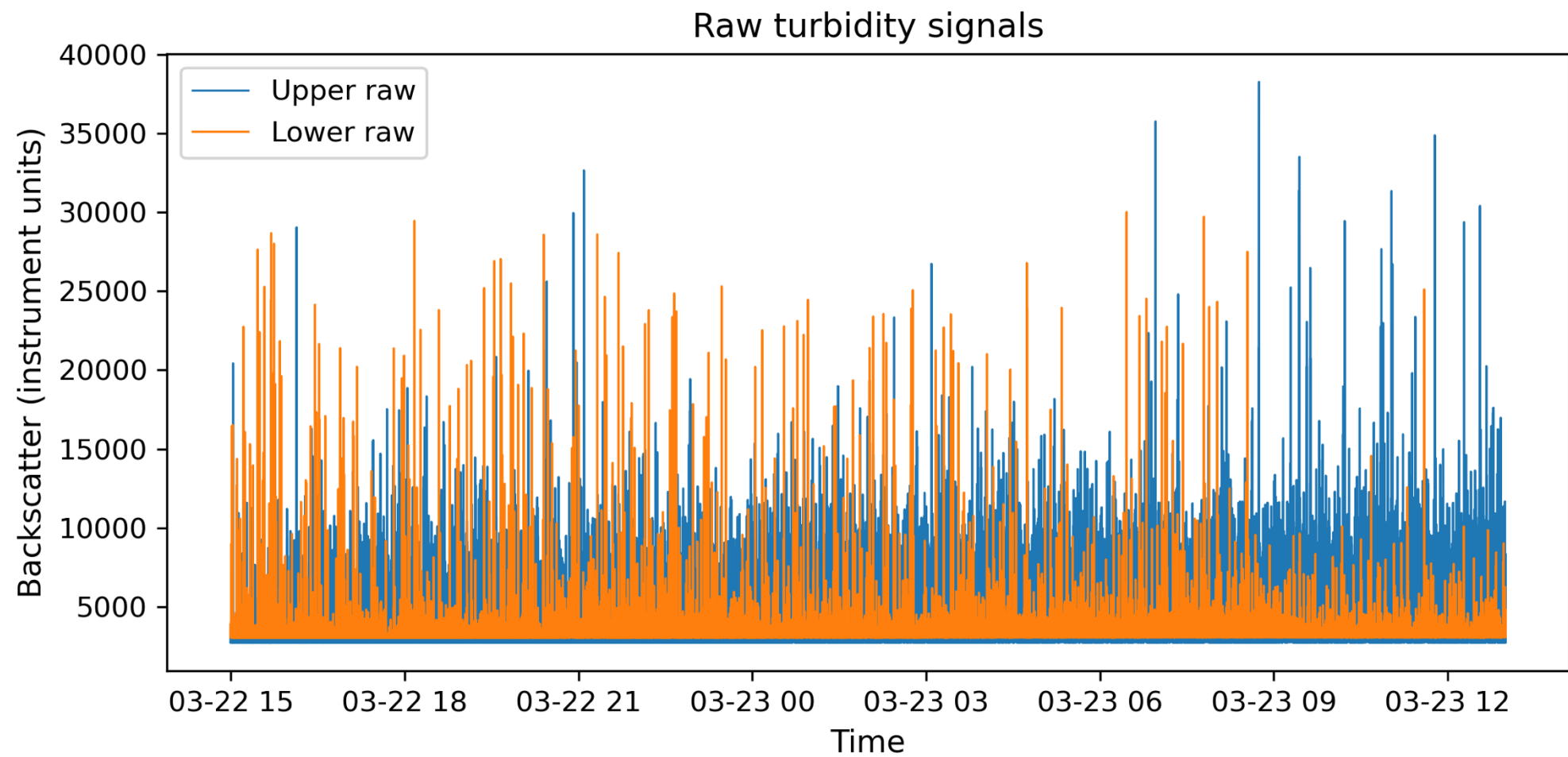
04/28/2026



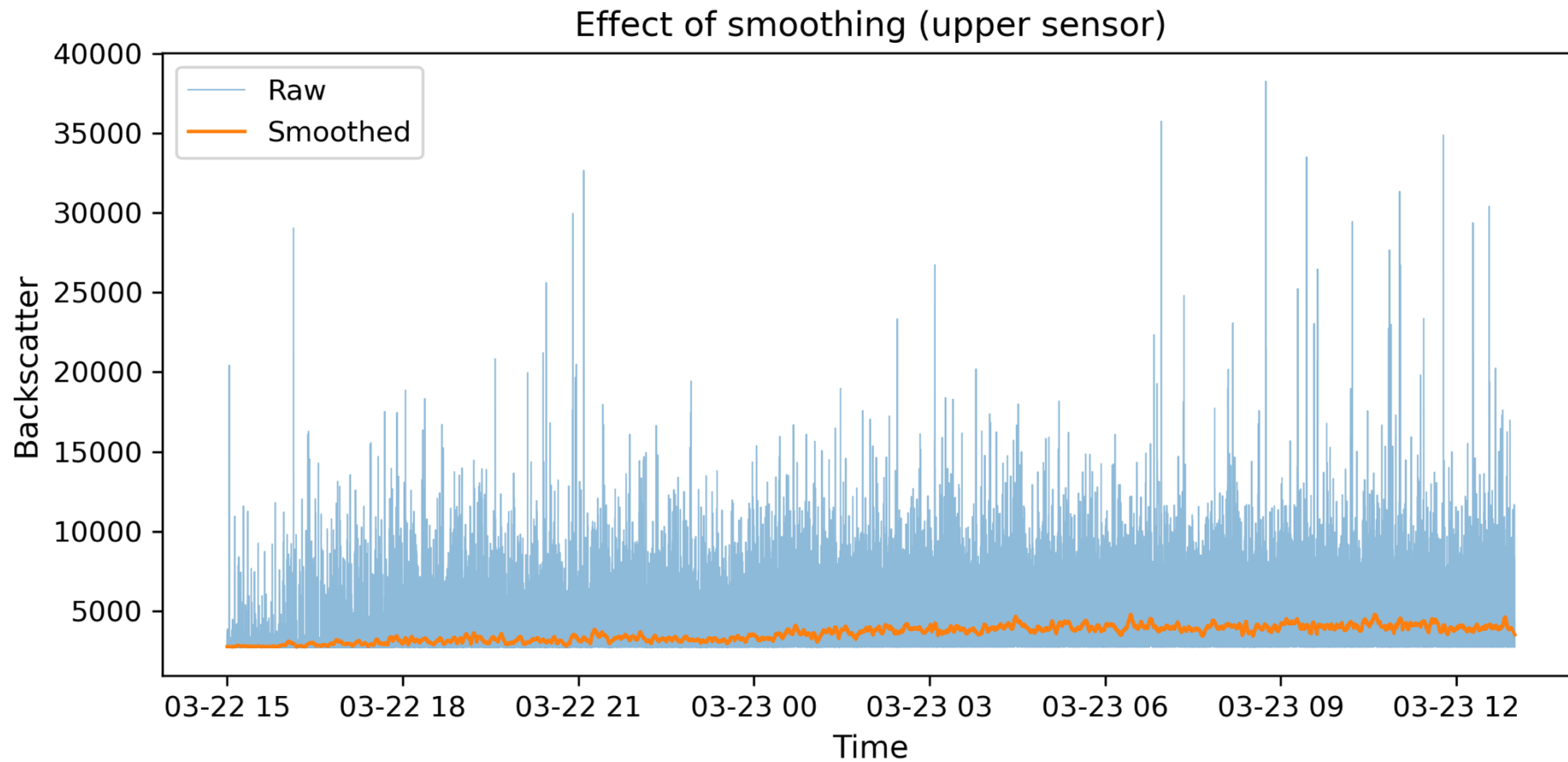
WHAT IS THE IMPACT OF
SINKING ASH ON OCEAN
CHEMISTRY?

ISAC

In Situ Autonomous
Column

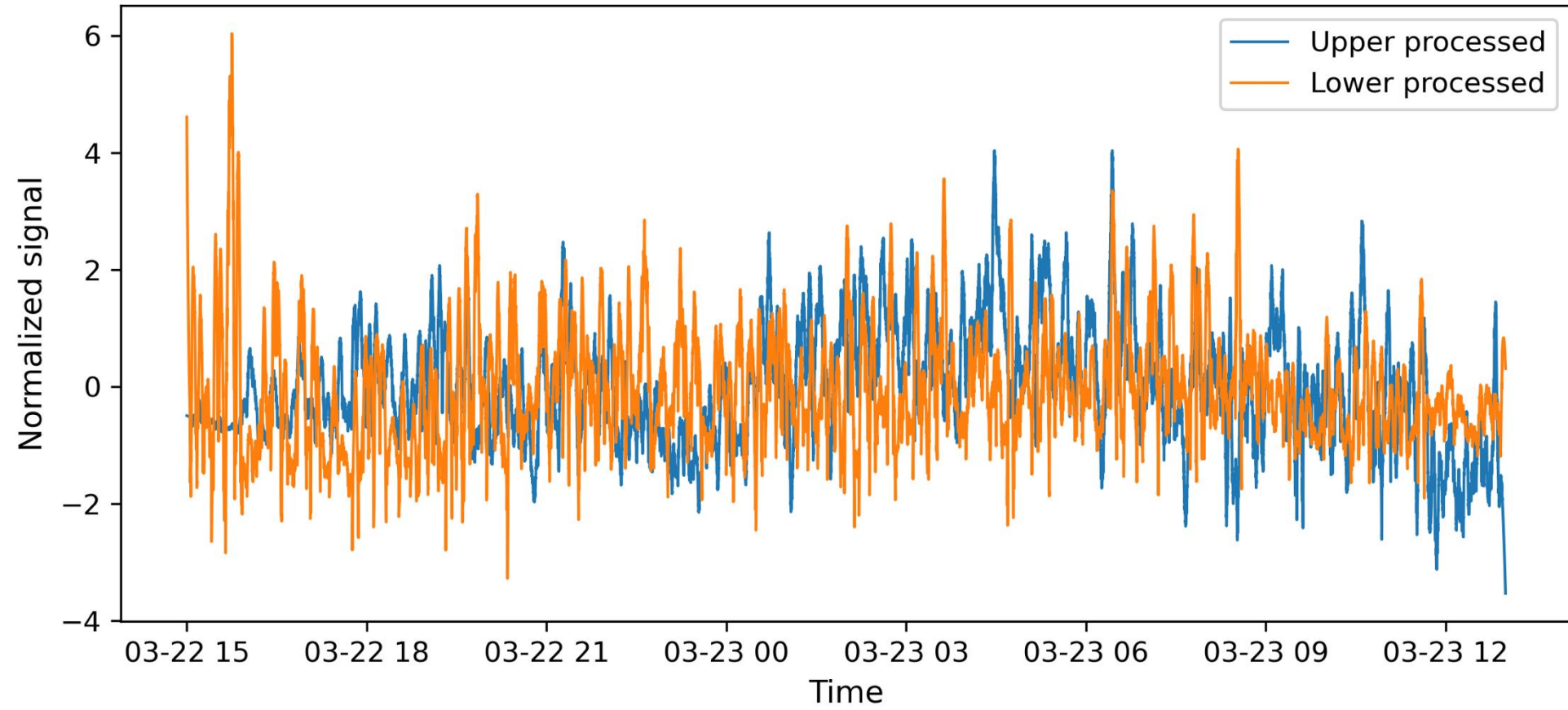


Distance between sensors: 2.7 m

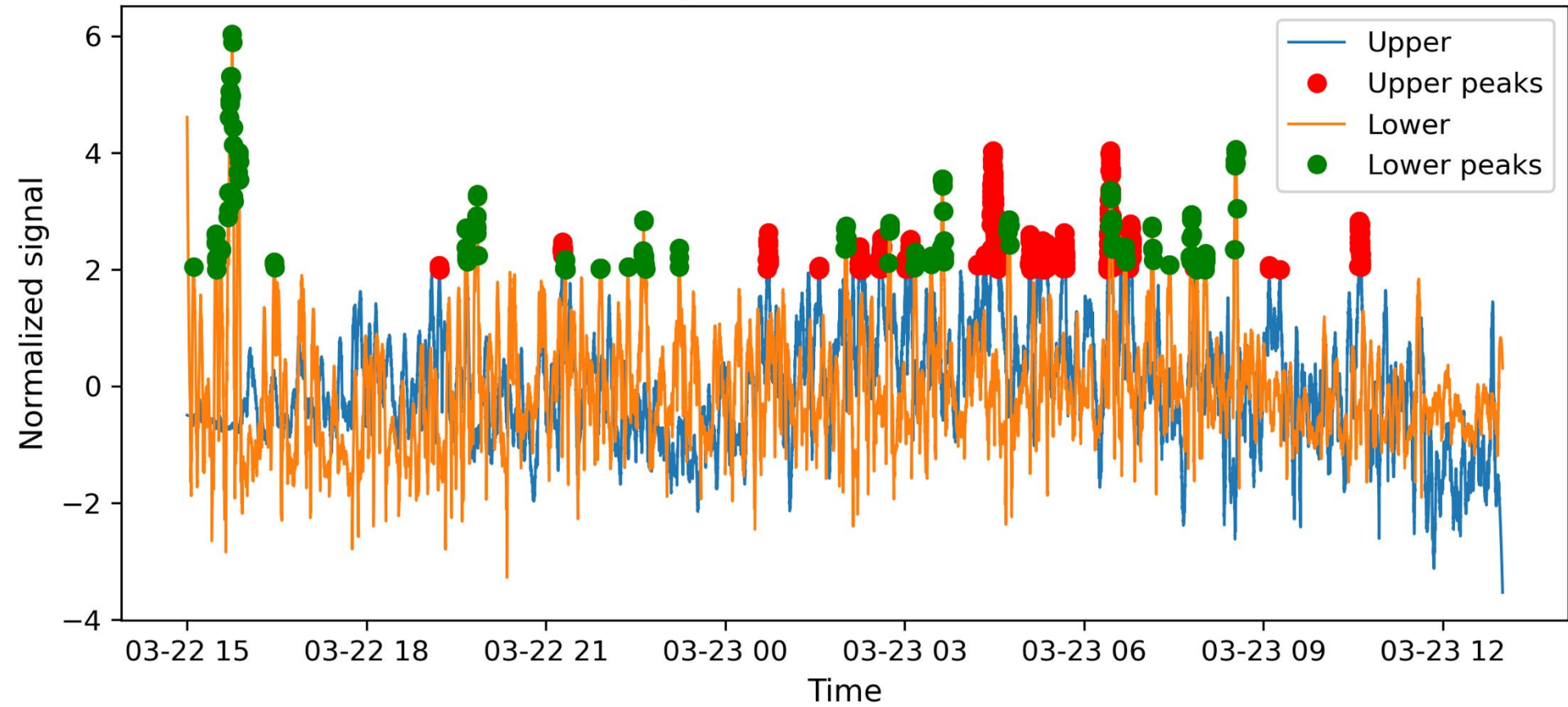


Uses a Savitzky-Gavol filter over 3 minutes; preserves peak shape (round) and timing, reduces noise

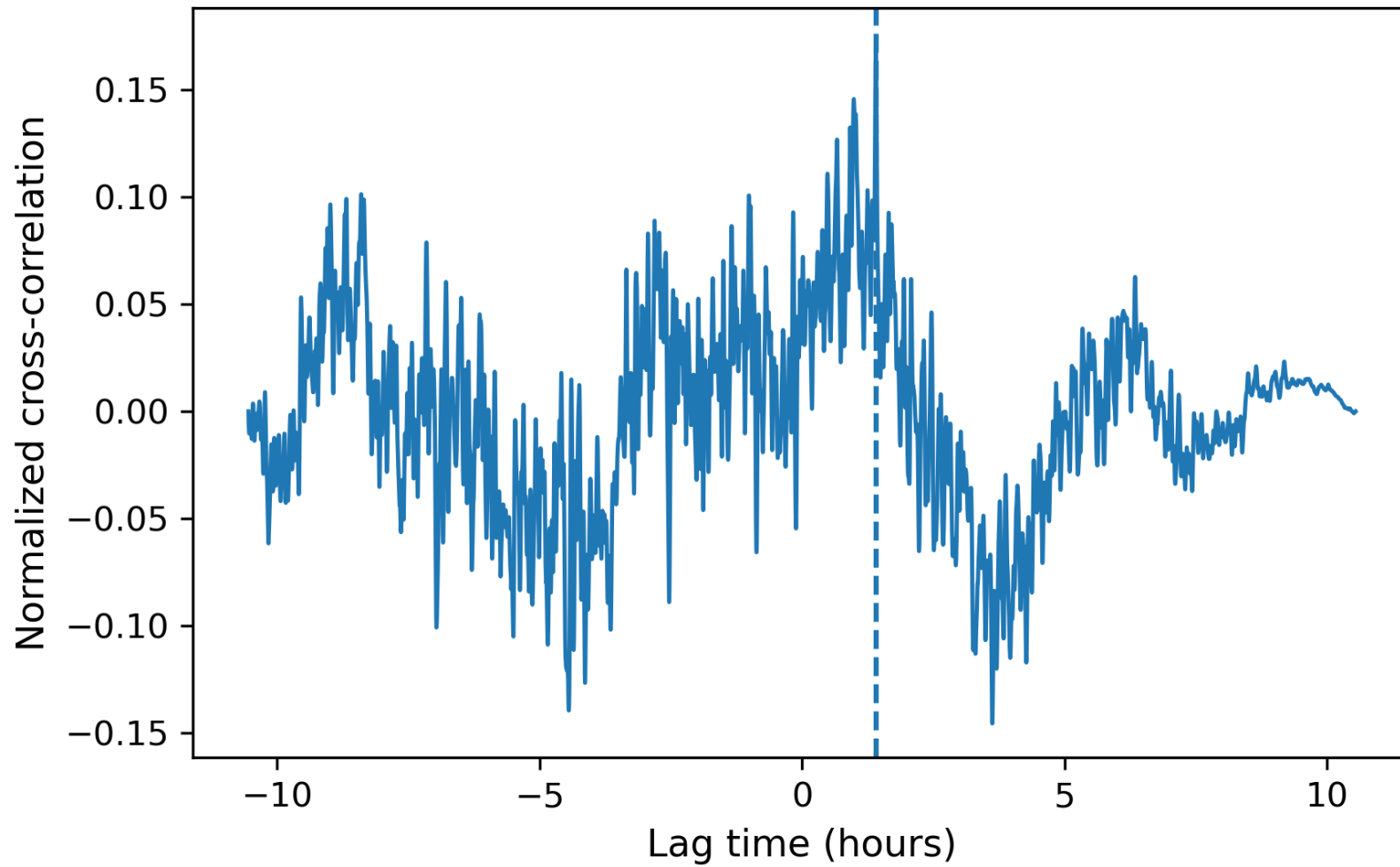
Signals used in correlation



Detected peaks used for lag estimation



Lag determination via cross-correlation



==== FINAL RESULT ====
Lag: 1.41 hours
Velocity: 46.02 ± 0.28 m/day

This result is robust across smoothing intervals.